

Арифметика значений синуса и косинуса

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Теоретический минимум.

Опр. A number is an *algebraic integer* if it is a root of a monic polynomial with integer coefficients; such numbers form a ring, and a rational algebraic integer is an ordinary integer. $2 \cos \frac{2\pi}{n} = \varepsilon + \varepsilon^{-1}$ is an algebraic integer of degree $\varphi(n)/2$.

Теорема 1 (Niven 1956) *If θ is a rational multiple of π and $\cos \theta \in \mathbb{Q}$, then $\cos \theta \in \{0, \pm \frac{1}{2}, \pm 1\}$. The same list, with $\pm \frac{\sqrt{2}}{2}$ excluded, describes the rational values of $\sin$ and, for $\tan$, gives $\{0, \pm 1\}$.*

Теорема 2 (Lindemann 1882) *π is transcendental; consequently $\sin r$, $\cos r$ and $\tan r$ are transcendental for every nonzero algebraic r , while $\cos \frac{2\pi}{n}$ is algebraic for every n .*

Факт 1 $T_n(\cos \theta) = \cos n\theta$ with $T_n \in \mathbb{Z}[x]$ turns any question about $\cos n\alpha$ into a question about iterating an integer polynomial — this is the engine of the whole sheet.

1. Prove Niven's theorem, and determine all triangles whose angles are rational multiples of π and whose cosines are all rational.
2. (Proposed problems) Suppose $r \in \mathbb{Q}$ and m is a positive integer with $\cos^m(r\pi) \in \mathbb{Q}$. Determine all possible values of $\cos r\pi$.
3. (MIPT 2012) Prove that if a polynomial with rational coefficients has the root $1 + \cos \frac{2\pi}{9} + \cos^2 \frac{2\pi}{9}$, then it also has the root $1 + \cos \frac{8\pi}{9} + \cos^2 \frac{8\pi}{9}$.
4. (Proposed problems) Let p be an odd prime and let $\alpha \in [0, \frac{\pi}{2}]$ satisfy $\cos \alpha = \frac{1}{p}$. Prove that for every integer $n > 1$ there is no integer m with $\cos(n\alpha) = \frac{1}{m}$.

5. Prove that $\frac{1}{\pi} \arccos \frac{1}{3}$ is irrational, and deduce that the dihedral angle of a regular tetrahedron is incommensurable with π . (*Подсказка: $\cos n\alpha = a_n/3^n$ с целым a_n , не делящимся на 3.*)